

Equivalent Delay of ZOH and Linear Interpolation

1 Time Domain: Impulse Response

For LTI reconstruction, output is convolution:

$$y(t) = \sum_k x[k] \cdot h(t - kT_s) \quad (1)$$

To find $h(t)$: set $x[0] = 1$, all other $x[k] = 0$, then $y(t) = h(t)$.

ZOH (hold for one period):

$$h_{\text{ZOH}}(t) = \begin{cases} 1, & 0 \leq t < T_s \\ 0, & \text{else} \end{cases} \quad (2)$$

Linear interpolation (output formula for $t \in [kT_s, (k+1)T_s)$):

$$y(t) = x[k-1] \cdot \frac{(k+1)T_s - t}{T_s} + x[k] \cdot \frac{t - kT_s}{T_s} \quad (3)$$

Setting $x[0] = 1$, others zero:

$$h_{\text{linear}}(t) = \begin{cases} t/T_s, & 0 \leq t < T_s \\ (2T_s - t)/T_s, & T_s \leq t < 2T_s \\ 0, & \text{else} \end{cases} \quad (4)$$

2 Laplace Transform

ZOH:

$$H_{\text{ZOH}}(s) = \int_0^{T_s} 1 \cdot e^{-st} dt = \frac{1 - e^{-sT_s}}{s} \quad (5)$$

Linear interpolation:

$$H_{\text{linear}}(s) = \int_0^{T_s} \frac{t}{T_s} e^{-st} dt + \int_{T_s}^{2T_s} \frac{2T_s - t}{T_s} e^{-st} dt = \frac{(1 - e^{-sT_s})^2}{s^2 T_s} \quad (6)$$

3 Frequency Response

Substitute $s = j\omega$:

ZOH:

$$H_{\text{ZOH}}(j\omega) = \frac{1 - e^{-j\omega T_s}}{j\omega} = e^{-j\omega T_s/2} \cdot \frac{2 \sin(\omega T_s/2)}{\omega} \quad (7)$$

Phase: $\angle H_{\text{ZOH}} = -\omega \cdot \frac{T_s}{2}$

Linear interpolation:

$$H_{\text{linear}}(j\omega) = \frac{(1 - e^{-j\omega T_s})^2}{(j\omega)^2 T_s} = e^{-j\omega T_s} \cdot \frac{4 \sin^2(\omega T_s/2)}{\omega^2 T_s} \quad (8)$$

Phase: $\angle H_{\text{linear}} = -\omega \cdot T_s$

4 Equivalent Delay

Phase delay relates to time delay by $\phi = -\omega\tau$, so:

Method	Delay τ	Delay Steps $d = \tau/T_s$
ZOH	$T_s/2$	0.5
Linear Interpolation	T_s	1.0